

Rao Faizan

Doctoral Researcher, Artificial Intelligence — Kyung Hee University, South Korea

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Summary

Doctoral researcher in artificial intelligence with five years of combined research and university teaching experience. Builds multi-agent (agentic) systems for cardiovascular care: clinical knowledge graphs constructed from raw registry data, hybrid retrieval-augmented reasoning over them, and the context-management, safety and alignment layers that make agent pipelines dependable. A parallel track shrinks clinical language models so they can be deployed for inference inside hospitals. Seeking research scientist or ML engineer roles in applied LLM, agentic and multimodal systems.

Technical Skills

Agentic Systems	Multi-agent orchestration, context management, tool-use and MCP interfaces, agent safety, alignment and guardrails, neuro-symbolic output verification
LLM & Efficiency	Fine-tuning (LoRA/PEFT), quantization, knowledge distillation, structured pruning; compressing clinical models for on-premises hospital inference
Knowledge & RAG	Knowledge-graph construction from raw clinical registries, schema design, link prediction, hybrid (dense + sparse + graph) retrieval-augmented generation
Multimodal & CV	Vision-language models, medical VQA and report generation (MIMIC-CXR), person re-identification, attention mechanisms, object detection
Time Series	LSTM and sequence models for demand forecasting, scheduling and optimization
Stack	Python (primary), C/C++, Java, SQL; PyTorch, Hugging Face Transformers, Keras/TensorFlow, scikit-learn, NumPy, pandas; Git, Linux, Docker, Jupyter

Research & Professional Experience

Doctoral Researcher — Agentic AI & Medical Knowledge Systems

Mar 2024 – Present

Kyung Hee University, Gyeonggi-do, South Korea

- Lead the multi-agent workstream of *Explainable Logical Reasoning for Medical Knowledge Generation*, a Korean government-funded programme: specialised agents for cardiovascular diagnosis, risk explanation and patient health literacy.
- Construct medical knowledge graphs from raw clinical registry data — the KorHF-III heart-failure registry (7,351 patients, 47 centres) became a 7,439-node, 114,505-edge graph — and query them through a HybridRAG pipeline combining graph traversal with dense and sparse retrieval.
- Design the context-management, safety and alignment layer: inter-agent context passing, scope limits on clinical claims, and a proof reasoner that verifies generated statements against a symbolic knowledge base before they reach a user.
- Shrink clinical models for hospital-side inference, including BANQ, a budget-aware nested quantization scheme serving one low-bit checkpoint across several memory budgets.
- Build vision-language models over the MIMIC chest X-ray corpus for clinical question answering and report generation; co-author grant proposals and project deliverables.

Lecturer — Department of Software Engineering

Oct 2022 – Feb 2024

University of Central Punjab (UCP), Lahore, Pakistan | [faculty profile](#)

- Taught seven undergraduate courses over three semesters, including Introduction to Machine Learning, Image Processing, Databases and Operating Systems.
- Supervised and examined final-year ML and computer vision projects; served on the “Become a Globalized University” strategic committee.

Lecturer — Department of Software Engineering

Sep 2021 – Nov 2022

University of Lahore (UOL), Lahore, Pakistan

- Taught Artificial Intelligence, Operating Systems, Human-Computer Interaction and Discrete Structures, Fall 2021 – Fall 2022; member of the Final-Year Project evaluation committee and organizer of workshops on $\text{\LaTeX}$ and full-stack web development.

Research Assistant — Machine Vision & Intelligent Systems Lab

Feb 2020 – Apr 2021

National University of Sciences & Technology (NUST), Islamabad, Pakistan

- Developed IAB-Net, an attention-based network for person re-identification across non-overlapping cameras; owned dataset curation, model design and ablation studies through to publication at ICoDT2 2021 (M.S. thesis).

Funded Research Projects

Explainable Logical Reasoning for Medical Knowledge Generation

2024 – Present

Kyung Hee University | IITP / Ministry of Science and ICT, Korea — IITP-2022-0-00078

- Multi-agent system for cardiovascular diagnosis support, grounded in knowledge graphs built from the KorHF-III registry and a clinician-curated chest-pain knowledge base, served through HybridRAG over a graph database with an MCP tool interface.
- Own the model-efficiency workstream (quantization, distillation, pruning), the agent safety/alignment layer, and the vision–language pipeline on MIMIC-CXR.

Hybrid AI-Based Multi-Agent System for Data-Driven Energy Operations

2024 – Present

Kyung Hee University / MSIT ITRC programme — DESIRe Center / Python, PyTorch, Keras

- Built a HybridRAG system over NREL open energy datasets so operational questions can be answered against heterogeneous grid and building-load data.
- Built the Self Demand Response (SDR) component: unified consumption datasets, trained the LSTM forecasting models driving the SDR policy, and designed the battery-storage scheduler.

Publications

16 citations, h-index 2 — [Google Scholar](#)

Journal Articles

- [J1] T. D. T. Nguyen, U. U. Rehman, M. Hussain, **R. Faizan**, J. Hussain, S.-H. Bae, J. U. Kim, S. T. Kim, and S. Lee. “Unified link prediction modeling for enhanced knowledge graph completion task.” *Expert Systems with Applications*, vol. 279, 127356, 2025. (*SCIE*)

Conference Papers

- [C1] C. Bold, **R. Faizan**, S. Lee, and S. T. Kim. “Automating medical multi-agent workflows from natural language using LLMs.” *Proc. Korea Computer Congress (KCC)*, Korean Institute of Information Scientists and Engineers, Jeju, Korea, Jun. 2026. (*Accepted*)
- [C2] C. Bold, **R. Faizan**, C.-H. Hua, S. M. Kang, S. Lee, and S. T. Kim. “Hybrid retrieval-augmented generation for chest-pain differential diagnosis: recall gains and calibration costs.” 2026. (*Accepted*)
- [C3] **R. Faizan**, S. Lee, and S. T. Kim. “ChestVQATrans: A transformer-based approach for medical VQA in radiology.” *Proc. 8th Artificial Intelligence and Cloud Computing Conference (AICCC)*, pp. 169–173, ACM, Dec. 2025. (*First author*)
- [C4] M. B. Yaseen, **R. Faizan**, M. Waseem, M. Kamran, and N. U. Haq. “Sarcasm detection in news headlines using deep learning with linguistic feature fusion.” *6th Int. Conf. on Innovative Computing (ICIC)*, pp. 1–6, IEEE, Dec. 2025.
- [C5] S. Wazir, **R. Faizan**, and D. Kim. “HistoSeg++: Delving deeper with attention and multiscale feature fusion for biomarker segmentation.” *Proc. 12th Int. Conf. on Biomedical and Bioinformatics Engineering (ICBBE)*, ACM, 2025. doi:10.1145/3794209.3794211
- [C6] **R. Faizan**, M. M. Fraz, and M. Shahzad. “IAB-Net: Informative and attention-based person re-identification.” *International Conference on Digital Futures and Transformative Technologies (ICoDT2)*, pp. 1–5, IEEE, 2021. (*First author*)

Conference Abstracts

- [A1] D. J. Choi, **R. Faizan**, C. Bold, S. Y. Lee, J. H. Kim, S. Jung, S. M. Kang, B. S. Yoo, H. J. Cho, J. C. Youn, J. J. Park, and M. Yoon. “Longitudinal trajectories of guideline-directed medical therapy and 12-month mortality in acute heart failure: insights from the KorHF-III Registry.” *ESC Congress 2026*, European Society of Cardiology, Munich, Aug. 2026. (*Accepted — abstract no. 87857*)

Education

Ph.D. in Computer Engineering — Artificial Intelligence

Mar 2024 – Present

Kyung Hee University (KHU), Gyeonggi-do, South Korea / CGPA 4.25 / 4.30 (99.33 percentile) / Presidential Scholarship

M.S. in Computer Science

Sep 2018 – Apr 2021

National University of Sciences & Technology (NUST), Islamabad, Pakistan / CGPA 3.2 / 4.0

Thesis: Informative Attention-Based Deep Network (IAB-Net) for Person Re-Identification

B.S. in Computer Science

Oct 2014 – Jun 2018

The Islamia University of Bahawalpur (IUB), Pakistan / CGPA 3.8 / 4.0 / 1st position, semesters 2–8

Honours, Awards & Certifications

- **Presidential Scholarship**, Kyung Hee University — full Ph.D. funding awarded for academic excellence, 2024–2028.
- **1st position, semesters 2–8**, The Islamia University of Bahawalpur — top of cohort in seven consecutive semesters, each carrying a merit scholarship, 2015–2018.
- **Certifications**: MIT CITI Data & Specimens Only Research (2024) · IBM Data Analysis and Machine Learning with Python (2023) · PIAIC Certified Cloud Native Developer (2023).